

6 | Conclusions and future developments

In this work, reinforcement learning was successfully applied to the air hockey task using a general-purpose manipulator. The agent was able to play entire matches without violating too many constraints, thanks to the ATACOM algorithm. The SAC method outperformed the rule-based controller for the hitting task, achieving a higher success rate. The rule-based switcher effectively selected the appropriate policy at the right time, with its parameters slightly optimized PGPE.

6.1. Future works

The developed rule-based switcher currently uses a few simple parametric rules. Future work could involve implementing a more complex switcher with additional rules that also take the opponent's actions into account.

An other strategy could be to apply an end-to-end hierarchical reinforcement learning approach, such as HIRO [16], where a single low-level policy receives from the high-level policy a vector to determine the desired behavior.

Since air hockey is a multi-agent setting, investigating a self-play learning strategy could also be beneficial. This approach could help the agent improve by continuously adapting to and learning from interactions with itself or other agents.

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